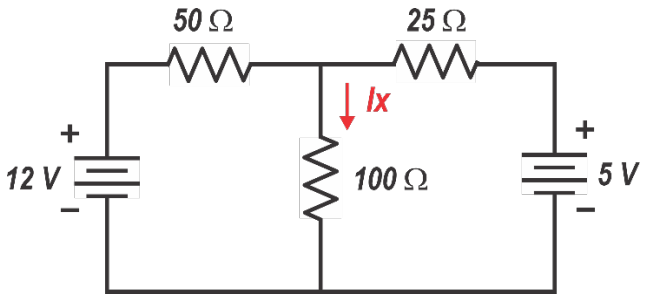
1. Nodal Analysis: Solve the nodal equations by hand, do **not** use a software solver (like Wolfram-Alpha).



A) **Assign voltage names to each node** in the circuit (include the ground reference). Draw the schematic and clearly label the node voltages.

B) What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$). You will be using this equation in (C).

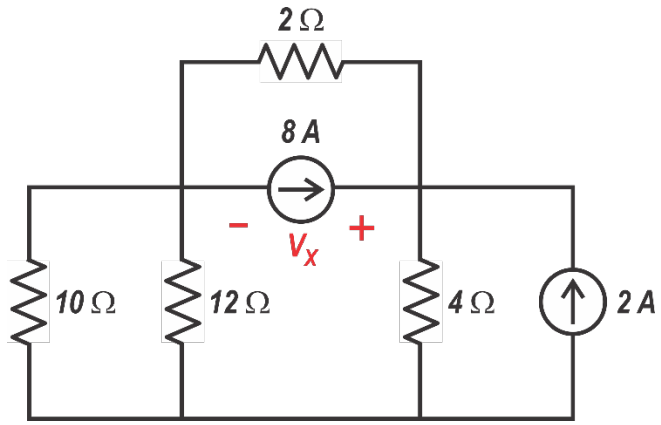
C) Write a sufficient **set of equations** to solve for all the node voltages. Clearly list your complete set of equations.

D) Solve the equations in (C) for the node voltages. Provide a **Table** of all the node voltage values.

E) Calculate I_x

F) Calculate the **power** in the two ***voltage*** sources. Determine whether each source is delivering or absorbing power, and briefly explain why.

2. Nodal Analysis: Solve the nodal equations by hand, do **not** use a software solver (like Wolfram-Alpha).



1. **Assign voltage names to each node** in the circuit (include the ground reference). Draw the schematic and clearly label the node voltages.

2. What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$). You will be using this equation in (C).

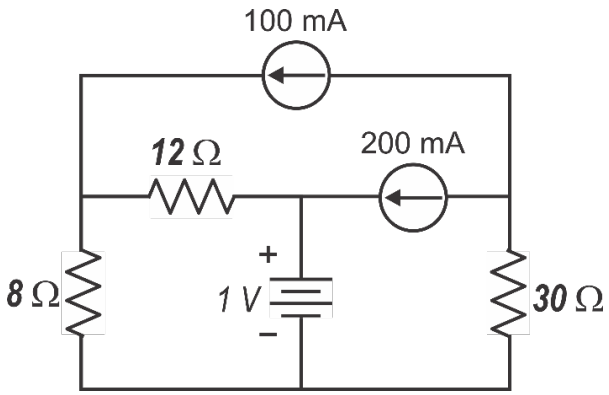
3. Write a sufficient **set of equations** to solve for all the node voltages. Clearly list your complete set of equations.

4. Solve the equations in (C) for the node voltages. Provide a **Table** of all the node voltage values.

5. Calculate V_x

6. Calculate the **power** in the two ***current*** sources. Determine whether each source is delivering or absorbing power, and briefly explain why.

3. Nodal Analysis: Solve the nodal equations by hand, do **not** use a software solver (like Wolfram-Alpha).



A) **Assign voltage names to each node** in the circuit (include the ground reference). Draw the schematic and clearly label the node voltages.

B) What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$). You will be using this equation in (C).

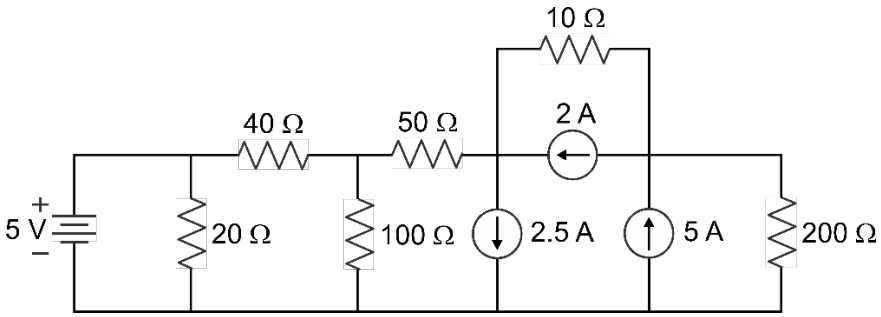
C) Write a sufficient **set of equations** to solve for all the node voltages. Clearly list your complete set of equations.

D) Solve the equations in (C) for the node voltages. Provide a **Table** of all the node voltage values.

E) Calculate the **power** in the ***voltage*** source. Determine whether the source is delivering or absorbing power, and briefly explain why.

F) Calculate the **power** for the two ***current*** sources. For each, determine whether the source is delivering or absorbing power, and briefly explain why.

4. Nodal Analysis: You **may** use a software solver (like Wolfram-Alpha).



A) **Assign voltage names to each node** in the circuit (include the ground reference). Draw the schematic and clearly label the node voltages.

B) What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$). You will be using this equation in (C).

C) Write a sufficient **set of equations** to solve for all the node voltages. Clearly list your complete set of equations.

D) Solve the equations in (C) for the node voltages. You **may** use a software solver, like Wolfram-Alpha, to find the unknown voltages. Provide a **Table** of all the node voltage values.